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AI-Based Network Intrusion Detection System

Project Report

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Table of Contents

1. Abstract.....	3
2. Introduction and Problem Statement.....	3
3. Literature Review.....	4,5
4. Methods/Model	5,6
5. Result.....	7
6. Discussion and Analysis.....	8
7. References	9

AI-Based Network Intrusion Detection System

1. Abstract

This project presents an AI-Based Network Intrusion Detection System (IDS) developed using machine learning and deep learning techniques to detect malicious network traffic. The system was built using the NSL-KDD dataset and treated intrusion detection as a binary classification problem, where network traffic is classified as either normal or an attack.

Several models were implemented, including Logistic Regression, Random Forest, and a custom Feedforward Neural Network developed in MATLAB. Data preprocessing techniques, such as numerical conversion and normalization, were applied before training to improve model performance.

The Neural Network used ReLU and Softmax activation functions with L2 regularization to reduce overfitting. The models were evaluated using accuracy, precision, recall, and F1-score. Experimental results showed that the Neural Network achieved the best performance with 86.7% accuracy and 87.5% F1-score, outperforming the other models.

The results demonstrate the effectiveness of artificial intelligence techniques in improving intrusion detection and enhancing cybersecurity systems.

2. Introduction and problem statement

With the increase in the number of networked systems, cybersecurity threats such as unauthorized access and network intrusion have also gained popularity. Intrusion Detection Systems (IDS) serve as one of the key solutions in detecting potential dangers and preventing any security breaches. Rule-based systems have proven to be less efficient in discovering new types of cyber attacks; therefore, the use of machine learning and deep learning algorithms becomes relevant when dealing with up-to-date intrusion detection problems.

The aim of the current work is to address the problem of detection of cyber intrusions through the use of machine learning and deep learning algorithms. More specifically, the objective is to develop a classifier that will help detect network intrusions by distinguishing between normal and malicious (attacking) network traffic. This will lead to solving a binary classification problem. For this project, the NSL-KDD database will be used. It is known to be one of the most popular databases used to solve intrusion detection problems. It consists of a large number of records representing either normal network traffic or a particular attack. The following features are included in the database: protocol type, service, connection duration, and other traffic characteristics.

Output for the classification algorithm would be the classification label, whereby 0 means normal traffic, and 1 means an attack. In order to find out the solution to this problem, several models will be used, beginning with the logistic regression, which acts as the initial baseline model, followed by the random forest, and lastly the deep learning model. In evaluating the models used, measures such as accuracy, classification report, which contains the measures of precision, recall, and F1-score, and the confusion matrix will be used.

3. Literature Review

Intrusion Detection Systems (IDS) play a critical role in modern cybersecurity by identifying malicious activities and protecting network infrastructure from evolving threats. With the increasing complexity of cyberattacks, traditional rule-based detection methods have become insufficient, leading to the adoption of machine learning (ML) and deep learning (DL) techniques for more accurate and adaptive intrusion detection. Early research in intrusion detection relied heavily on signature-based and anomaly-based systems. While signature-based approaches are effective in detecting known attacks, they fail to identify new or unknown threats. Anomaly-based methods attempt to address this limitation by modeling normal behavior and detecting deviations. These systems often suffer from high false positive rates, which limit their practical deployment. As a result, researchers began integrating machine learning techniques to improve detection, performance, and adaptability.

Machine learning algorithms such as logistic regression, support vector machines (SVM), and random forest have been widely used in IDS research. Logistic regression serves as a baseline model due to its simplicity and interpretability, but its performance is often limited when dealing with complex nonlinear data patterns. In contrast, ensemble methods like random forest have demonstrated superior performance by combining multiple decision trees to improve classification accuracy and robustness. Studies have shown that random forest is practically effective when applied to structured data sets such as NSL-KDD, as it can handle high-dimensional data and capture features in interactions more efficiently.

More recently, deep learning approaches have gained attention due to their ability to automatically extract complex patterns from data. Neural networks have been applied to intrusion detection tasks to improve classification performance. These models can learn hierarchical representations of network traffic, making them suitable for detecting sophisticated attack patterns. However, deep learning models require careful tuning to avoid overfitting, especially when dealing with imbalanced data sets such as NSL-KDD, where certain attack categories are underrepresented.

The NSL-KDD dataset remains one of the most used benchmarks for evaluating IDS models. It was introduced as an improved version of the KDD Cup 1999 dataset to address issues such as redundancy and bias. Despite its age, NSL-KDD continues to be widely used in academic research due to its structured format and predefined training and testing splits. Researchers commonly utilize binary classification approaches on this dataset, where network traffic is categorized as either normal or malicious. This approach simplifies the detection process while still providing effective identification of intrusions.

Several studies have compared machine learning and deep learning approaches on NSL-KDD and similar datasets. Results generally indicate that ensemble methods like random forest achieve high accuracy due to their robustness, while neural networks offer competitive performance when properly regularized. However, the effectiveness of these models is influenced by factors such as data pre-processing, feature scaling, class imbalance, and model tuning techniques, such as normalization, early stopping, and regularization have been shown to significantly improve generalization performance.

In summary, the literature demonstrates that no single model is universally optimal for intrusion detection instead, performance depends on the dataset's characteristics and the chosen evaluation approach. Combining multiple models and applying proper pre-processing techniques can lead to more reliable and realistic results. This project builds upon these findings by comparing baseline, ensemble and deep learning models on the NSL-KDD dataset using a Binary classification approach with a focus on achieving balanced and generalizable performance.

4. Methods/ Model

Before training the models, the NSL-KDD data was prepared and cleaned to improve data quality and enhance model performance during learning. Textual values such as protocol type, service type, and connection state were converted to numerical values so the models could handle them correctly. The data was also reviewed to ensure there were no duplicates or inconsistencies that could affect the results. Normalization and property standardization were then applied to the numerical data to minimize large discrepancies between different values and make the learning process more balanced and stable. These steps significantly improved model accuracy and accelerated the training process.

Logistic Regression Model Methodology:

The baseline used was logistic regression for the task of classification into either normal or attack traffic, using the NSL- KDD data set. Logistic regression was chosen because it is a straightforward approach that has been used widely in benchmarking intrusion detection systems.

Preprocessing was carried out on the raw data to enhance its effectiveness in improving performance in several ways. First, categorical attributes were encoded in numeric form to be usable by machine learning algorithms. Attributes with extremely sparse values were also excluded since they only added noise to the data. In addition, all attribute values were normalized, which they had their ranges changed to 0 to 1. so as to come up with the final version of the model, separate data sets for training and test were used as opposed to using one dataset for both training and testing. This made the evaluation more practical. Other approaches tried included using various combinations of preprocessing procedures to achieve improved accuracy.

Random Forest Model Methodology:

The Random Forest model was used for the task of classification into either normal or attack traffic using the NSL-KDD data set. Random Forest was chosen because it is a powerful ensemble learning approach that combines the predictions of multiple trees to improve classification performance and reduce overfitting. This approach has been widely used in intrusion detection systems.

Preprocessing was carried out on the raw data to enhance its effectiveness in improving performance in several ways. First, categorical attributes were encoded in numeric form to be usable by machine learning algorithms. Attributes with extremely sparse values were also excluded since they only added noise to the data. To arrive at the final version of the model, separate data sets for training and test were used as opposed to using one dataset for both training and testing. This made the evaluation more practical. Other approaches, such as using different preprocessing combinations and parameter settings, were also tested, but some of them did not provide significant improvements in accuracy.

The Random Forest model was implemented in MATLAB using the TreeBagger function. The model was trained using multiple trees, where each tree was built using different subsets of the training data, and the final prediction was determined based on the majority vote of all trees. The resulting classifications were evaluated using confusion matrices as well as precision, recall, and F1-score for comparison with the other models.

Neural Network Model Methodology:

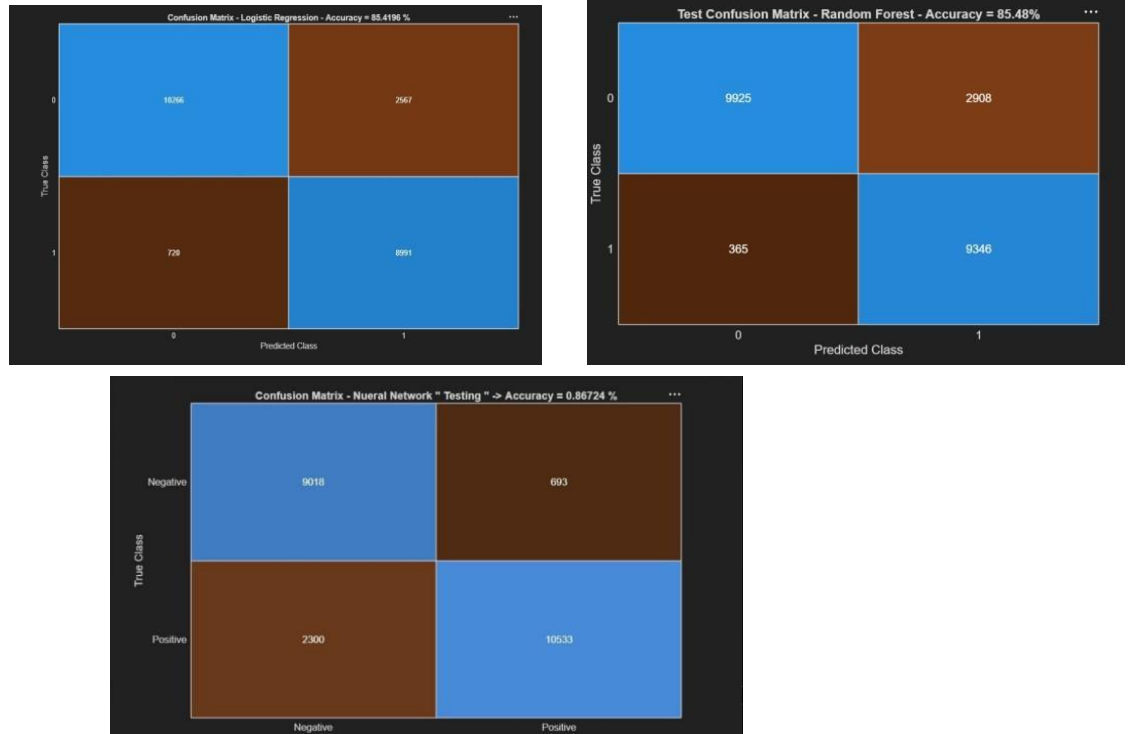
The proposed intrusion detection system (IDS) was implemented using a custom feedforward neural network developed in MATLAB. The model was designed for binary classification, where network traffic records are categorized as either normal or abnormal. The NSL-KDD dataset was used for both training and testing purposes after pre-processing and normalization. The neural network architecture consists of an input layer corresponding to the dataset features, two hidden layers containing 16 and 8 neurons, and an output layer with two neurons representing the binary classes.

The Rectified Linear Unit (ReLU) activation function was used in the hidden layers due to its computational efficiency and ability to reduce vanishing gradient problems. In the output layer, the SoftMax function was applied to generate probability distributions for the two classes. To improve model generalization and reduce overfitting, L2 regularization was implemented during training. Model parameters were optimized using forward propagation and backward propagation with gradient descent weight updates.

The dataset was processed over 1050 epochs using a learning rate of 0.1. During each epoch, random samples were selected from the training data to improve learning diversity and reduce memorization. Cross-entropy loss was used to evaluate prediction error during training.

Finally, the trained weights and biases were saved and reused during the testing phase to evaluate the model on unseen NSL-KDD test data.

5. Results



	Logistic Regression	Random Forest	Neural Network
Accuracy	85.42%	85.48%	86.72%
Precision	0.9345	0.96453	0.9383
Recall	0.8000	0.7734	0.8208
F1-Score	0.8620	0.85845	0.8756

The experimental results demonstrate that all three models achieved strong performance on the NSL-KDD binary classification task. The Neural Network achieves the highest overall accuracy (86.72%) and a F1-score (0.8756), indicating better generalization and intrusion detection capability compared to Logistic Regression and Random Forest. Precision, recall, and F1-score were used as the primary evaluation metrics to measure classification effectiveness and false alarm behavior. Overall, the obtained results align with the expectations and are consistent with prior studies showing that Neural Networks can provide improved performance for intrusion detection systems.

6. Discussion and Analysis

The experimental results showed that all three machine learning models were capable of detecting malicious network traffic with relatively high performance on the NSL-KDD dataset. Among the evaluated models, the Neural Network achieved the best overall performance, obtaining the highest accuracy and F1-score. This indicates that deep learning models are more effective at learning complex patterns and relationships within network traffic data compared to traditional machine learning approaches such as Logistic Regression and Random Forest.

The confusion metrics demonstrated that the models were generally successful in distinguishing between normal and abnormal traffic; however, some attack samples were still misclassified as normal traffic. This highlights one of the main limitations of intrusion detection systems, where certain attacks may share similar characteristics with legitimate network behavior. Additionally, the NSL-KDD dataset contains class imbalance and limited real-world variability, which may affect the model's ability to generalize perfectly to unseen environments.

Another limitation of this approach is that the implemented Neural Network required careful tuning of hyper parameters, such as the learning rate, number of epochs, and regularization strength. During experimentation several configurations resulted in overfitting where the model achieved very high training accuracy but weaker testing performance. Techniques such as normalization, L2 regularization, and control training epochs were therefore necessary to improve generalization.

The results also provided valuable insights into the behavior of machine learning models in cybersecurity applications, simpler models such as Logistic Regression produced stable, and interpretable results, while the Neural Network demonstrated stronger learning capability at the cost of increased computational complexity and tuning requirements. Future improvements could include using larger and more modern intrusion detection datasets, implementing advanced deep learning architectures, or applying additional techniques, such as dropout and feature selection to further improve detection accuracy and robustness.

9. References

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